# AATIF: A Multiplicative Governance Equation for Arabic-First LLM Safety

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**Abstract**

We present AATIF, a mathematical governance framework for large language model safety that provides an explicit, interpretable safety equation with non-compensability by construction. The core governance function $S = \sigma(w_1 \cdot I + w_2 \cdot E) \cdot [1 - \sigma(\alpha(H - \theta(d)))]$ fuses three independent perception channels—semantic harm proximity ($H$), intent classification ($I$), and emotional load ($E$)—through a multiplicative gate structure that prevents benign intent from compensating for detected harm. Unlike trained classifiers, AATIF requires zero training data, operating through 249 curated semantic anchors ($H$=171, $I$=46, $E$=32) with cosine similarity scoring via multilingual embeddings (bge-m3). The harm threshold $\theta$ is domain-parameterized—$\theta(d)$ ranges from 0.25 (healthcare) to 0.50 (creative)—allowing the same system to be stricter where consequences are severe and more permissive where expression freedom matters. The framework treats Arabic as a first-class semantic input, with 31 dialect-hyperbole disambiguation anchors to prevent figurative speech from triggering false harm signals. Beyond the safety equation, AATIF implements a full governance pipeline: $S(d)$ (safety scoring) $\rightarrow$ $P(d)$ (domain-specific deterministic protocols) $\rightarrow$ $R(d)$ (response style adaptation) $\rightarrow$ LLM generation $\rightarrow$ Output Gate (final safety and quality verification), orchestrated by a Governor module that provides a complete audit trail. Evaluated against HarmBench (236 behaviors) and MultiJail (75 prompts), AATIF achieves 74.3% safety-category detection with 100% on chemical/biological threats, and Arabic outperforms English on identical harmful prompts (74.7% vs 69.3%), validating the Arabic-first design. The system includes hysteresis control for decision stability, unconditional CBRN safety gates, and four tunable mode profiles calibrated via A/B testing (in-sample $F_1 = 0.984$ on 54 cases; held-out $F_1 = 0.9615$ on 56 unseen cases after anchor refinement). A test suite of 790 deterministic tests ensures mathematical correctness across all modules. AATIF combines, for the first time in a single deployable system, a closed-form multiplicative governance equation with non-compensability by construction (supplemented by an external hard override for absolute high-harm refusal), Arabic dialect-aware anchor scoring, and zero-fine-tuning extensibility, designed for human-over-the-loop governance.

**Keywords:** LLM safety, governance equation, Arabic NLP, multiplicative gating, non-compensability, semantic anchors

## 1 Introduction

Large language model (LLM) safety systems share three structural limitations that persist across both commercial platforms and academic proposals.

**Gap 1: No interpretable equation.** No deployed safety system provides an explicit, interpretable mathematical governance formula with provable properties. FanarGuard [Fatehkia et al., 2025] is a 2B-parameter classifier trained on 468K prompt-response pairs; Llama Guard 3

[Inan et al., 2024] is a fine-tuned language model performing binary classification; Google's Perspective API [Google Jigsaw, 2017] returns per-attribute toxicity scores from a neural classifier. All three produce scores, but none exposes a closed-form equation that a reviewer can inspect, verify, or prove properties about. The gap is not one of performance but of *interpretability*: when a safety system refuses a request, why?

**Gap 2: No non-compensability guarantee.** Existing aggregation methods—typically additive or mean-based—allow benign signals in one channel to compensate for harm detected in another. A prompt expressing harmful intent in calm, polite language can score as "moderate risk" under additive aggregation, because the low emotional-load score partially offsets the high harm score. This *toxic positivity* attack vector is a structural property of additive scoring, not an implementation bug.

**Gap 3: Arabic as afterthought.** Arabic dialect awareness is either absent or retrofitted onto English-first architectures. Arabic colloquial speech routinely uses violent metaphors for everyday emotions—*wallāhi ba-dhbaḥuh* ("I swear I'll slaughter him") means *I'm annoyed at him*—and English-first embedding spaces collapse these figurative expressions into their literal semantic neighborhoods, producing systematic false positives. No existing system treats Arabic dialectal hyperbole as a first-class safety concern.

This paper presents AATIF (Architected Adaptive Thoughts & Intelligence Frameworks), a governance framework that addresses all three gaps with a single mathematical structure. The core contribution is the $S$ *equation*:

$$S = \sigma(w_1 \cdot I + w_2 \cdot E) \cdot [1 - \sigma(\alpha(H - \theta(d)))] \tag{1}$$

where $H$ is semantic harm proximity, $I$ is intent classification, $E$ is emotional load, $\sigma$ is the logistic sigmoid, and $\theta(d)$ is a domain-parameterized harm threshold (Section 3.7). The multiplicative structure ensures *non-compensability*: when $H$ exceeds $\theta$, the harm gate drives $S$ toward zero regardless of how benign the intent or calm the emotional state. A hard override ($H > 0.7 \Rightarrow$ SAFE_FREEZE) provides an absolute safety floor independent of parameter calibration.

AATIF originated from 73 governance conjectures derived through a year of sustained observation across four LLM platforms (ChatGPT, Claude, Gemini, DeepSeek), documented in 78 field notes available as supplementary material. This paper operationalizes one of those conjectures—Conjecture #063, that governance operates through probabilistic constraint density rather than rule retrieval—into a computational engine with an explicit mathematical formulation, a comprehensive test suite of 790 deterministic tests, and benchmark evaluation against HarmBench and MultiJail.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 presents the S equation, its three scorers, the hysteresis controller, safety gates, mode profiles, domain parameterization, and the full governance pipeline ($S(d) \to P(d) \to R(d) \to$ LLM $\to$ Output Gate). Section 4 describes the Arabic-first design. Section 5 reports experimental evaluation. Section 6 discusses contributions and limitations. Section 7 concludes.

# 2 Related Work

## 2.1 Alignment Approaches

The dominant paradigm in LLM alignment treats governance as top-down principle application. Christiano et al. [2017] established reinforcement learning from human feedback (RLHF), learning reward functions from pairwise preference comparisons. Ouyang et al. [2022] applied this at scale in InstructGPT, demonstrating that preference-based fine-tuning shifts model outputs toward human-annotated intent. Bai et al. [2022] proposed Constitutional AI, in which natural-language principles guide self-critique and revision. In all three approaches, governance principles precede observation: the direction of derivation runs from principle to model.

A parallel body of work has examined how alignment operates internally. Arditi et al. [2024] demonstrated that refusal behavior across thirteen chat models is mediated by a single linear direction in activation space, rather than by distributed rule-following mechanisms. Lee et al. [2024] showed that direct preference optimization (DPO) does not eliminate harmful capabilities but learns to route around them, with the original capacity remaining recoverable. Wolf et al. [2024] formalized this fragility through Behavior Expectation Bounds, proving that any alignment process that attenuates rather than removes a behavior remains susceptible to adversarial elicitation. Taken together, these findings suggest that alignment functions as probabilistic constraint density on output distributions rather than as explicit rule retrieval—a characterization that motivates AATIF's mathematical approach to governance.

## 2.2 Arabic-Aware AI Safety

FanarGuard [Fatehkia et al., 2025] is the most directly relevant system: a bilingual Arabic-English content moderation classifier developed by the Qatar Center for Artificial Intelligence, trained on 468K prompt-response pairs and fine-tuned into a 2B-parameter model achieving $F_1 = 0.82$ on Arabic safety classification. FanarGuard introduces a dual-axis evaluation—safety (harmful→harmless) and cultural alignment (aligned→unaligned)—representing an important advance in culturally-aware moderation. Llama Guard 3 [Inan et al., 2024] extends Meta's safety classifier to eight languages including Arabic, applying the MLCommons 13-category taxonomy through binary classification. Google's Perspective API [Google Jigsaw, 2017] provides per-attribute toxicity scores with Arabic language support. Each of these systems operates as a trained classifier or fine-tuned language model; none provides an explicit, interpretable governance equation.

In Arabic hate speech detection, ADHAR [Aldjanabi et al., 2024] is a multi-dialectal corpus covering five Arabic variants (Egyptian, Levantine, Gulf, Maghrebi, MSA) with inter-annotator agreement $\kappa = 0.95$; cross-dialect experiments show accuracy dropping from 0.84 to 0.64 when training on one dialect and testing on another, empirically demonstrating that dialect-awareness is not optional for Arabic safety systems. SOD [Asiri & Saleh, 2024] contributes a Saudi-specific offensive language corpus of 24K+ tweets with hierarchical annotation, achieving $F_1 = 0.87$ with AraBERT. Both datasets informed AATIF's anchor expansion for dialectal coverage.

AATIF differs from these systems in three structural respects: (1) it provides an explicit mathematical equation with structural properties rather than learned classifier weights, (2) its multiplicative gate prevents intent from compensating for harm—non-compensability by construction that additive aggregation cannot provide, and (3) it requires zero training data, operating through curated semantic anchors. These differences position AATIF as complementary to data-driven classifiers rather than competitive with them.

## 2.3 Safety Benchmarks

Two benchmark suites are relevant to evaluating AATIF. HarmBench [Mazeika et al., 2024] provides 236 harmful behaviors across seven semantic categories (chemical/biological, cybercrime, illegal activities, harassment, harmful content, misinformation, copyright). HarmBench is English-only, creating a language mismatch with AATIF's Arabic-first anchor design that tests cross-lingual transfer. MultiJail [Deng et al., 2024] addresses the multilingual gap with 75 harmful prompts translated by native speakers into 10 languages including Arabic, enabling direct comparison of detection performance across languages. The MLCommons AI Safety taxonomy [MLCommons, 2024] provides a standardized 13-category harm classification (S1–S13) serving as a common reference for coverage comparison.

# 3 The AATIF Governance Equation

## 3.1 Definition and Properties

The AATIF governance engine computes a composite safety score $S \in [0, 1]$ from three independent perceptual channels:

$$S = \sigma(w_1 \cdot I + w_2 \cdot E) \cdot \underbrace{[1 - \sigma(\alpha(H - \theta(d)))]}_{\text{harm gate}} \tag{2}$$

where $\theta(d)$ is a domain-parameterized harm threshold (Section 3.7). This equation is a designed structure, not an empirically discovered law. Its form was chosen so that a single desirable property—non-compensability between harm and the other channels—holds by construction. We describe the structure, state precisely what it does and does not guarantee, and treat its parameters as hand-set design choices pending calibration.

**Quality term.** $\sigma(w_1 \cdot I + w_2 \cdot E) \in (0, 1)$ assesses how clear and benign the request is. Intent $I$ captures harmful-planning signals (higher $I$ = more benign); emotional load $E$ captures distress or escalation markers (higher $E$ = calmer). The two are combined additively inside the sigmoid, by deliberate contrast with the harm term: intent and emotion are treated as mutually substitutable evidence about request quality (strong benign intent can reasonably offset mild distress), whereas harm is not allowed to be substitutable at all. The weights $w_1 = 2.0$ and $w_2 = 1.5$ encode the design judgment that intent matters more than emotional state; these are hand-set values, not optimized parameters, and Section 6.2 lists their calibration as future work.

**Harm gate.** $[1 - \sigma(\alpha(H - \theta(d)))] \in (0, 1)$ is a soft gate controlled by semantic harm proximity $H$. The steepness $\alpha = 10$ controls transition sharpness; the threshold $\theta(d)$ sets where the gate begins to close and is parameterized by the application domain $d$ (Section 3.7), with a default of $\theta = 0.40$ for general use. When $H \ll \theta(d)$ the gate is open ($\approx 1$); when $H \gg \theta(d)$ it closes ($\approx 0$), driving $S$ toward zero.

**Non-compensability (what the structure guarantees).** Because $S$ is the product of the quality term and the harm gate, a rising $H$ suppresses $S$ multiplicatively regardless of how high $I$ or $E$ are: no combination of benign intent and calm emotion can restore $S$ once the harm gate is closing. This is the property additive aggregation ($S = aH + bI + cE$) lacks, where sufficiently high $I$ and $E$ always offset moderate $H$. We state the property precisely: as $H$ increases, the gate term decreases monotonically toward 0, so $S$ is bounded above by the gate term and is driven arbitrarily close to 0. Both factors are strictly positive (the sigmoid never reaches 0 or 1), so $S$ never equals exactly 0 by the equation alone—the gate produces strong attenuation, not a mathematical zero.

**The absolute floor is external, not equational.** Because the equation alone cannot force $S$ to exactly 0, the absolute safety guarantee is provided outside the equation by a hard override: when $H > 0.7$, the system triggers SAFE_FREEZE regardless of $S$. This override is imposed externally and is not derived from the equation; it ensures no parameter tuning can lift output above the safety floor in the high-harm regime. We are explicit about this division of labor: the equation provides interpretable, monotonic gating; the hard override provides the absolute guarantee. Claims of "guaranteed" refusal in the high-harm regime rest on the override, not on the closed form.

**Sigmoid symmetry.** The harm gate satisfies $[1 - \sigma(\alpha(H - \theta(d)))] = \sigma(\alpha(\theta(d) - H))$, verified formally, ensuring the gate is centered exactly at $\theta(d)$ and monotonically decreasing in $H$.

**Decision mapping.** The continuous score $S$ maps to four discrete actions:

Table 1: Decision mapping from $S$ score to governance action.

| Condition | Action | Behavior |
|---|---|---|
| $S > 0.7$ | EXECUTE | Process request normally |
| $0.5 < S \leq 0.7$ | CLARIFY | Request clarification before proceeding |
| $0.3 < S \leq 0.5$ | SAFE_STOP | Decline with explanation |
| $S \leq 0.3$ | SAFE_FREEZE | Maximum caution (score-based) |
| $H > 0.7$ | SAFE_FREEZE | Unconditional refusal (hard override) |

**Follow-up signal.** A follow-up signal $F$ provides a harm-floor guarantee: even when $S$ is high, $F$ flags residual risk for downstream monitoring, ensuring that borderline cases receive continued attention.

### 3.2 The Three Scorers

Each scorer operates independently, producing a continuous score in $[0, 1]$ through cosine similarity against curated semantic anchors using multilingual embeddings (bge-m3 via Ollama). A TF-IDF character n-gram fallback is available for offline operation without GPU.

**H scorer—Semantic Harm Proximity (*harārat al-kalima*).** The H scorer maintains 171 curated anchors spanning violence, weapons, self-harm, cybercrime, hate speech, and chemical/biological categories. Each anchor is a short phrase representing a harm concept. Input text is embedded via bge-m3 and compared against all anchors via cosine similarity. *Top-K aggregation* ($K = 3$) averages the three highest-similarity scores, preventing any single anchor from dominating the result. This mitigates the *anchor contamination* effect, in which one high-similarity anchor inflates the score despite low agreement across the anchor set. The H scorer includes 31 dialect-specific hyperbole anchors scored at $H \leq 0.05$, providing semantic "counter-gravity" for figurative Arabic expressions (Section 4).

**I scorer—Intent Classification (*al-niyya*).** The I scorer maintains 46 intent anchors organized into categories: informational, transactional, navigational, creative, and harmful-planning. Input is classified via top-$K$ softmax over anchor similarities, with an out-of-distribution (OOD) guard that flags inputs dissimilar to all intent categories. Confidence bands map similarity scores to discrete confidence levels.

**E scorer—Emotional Load (*al-shu'ūr*).** The E scorer maintains 32 emotion anchors covering distress, escalation, calmness, and neutral states. Higher $E$ indicates calmer emotional state; distress and escalation indicators produce lower $E$ scores, which reduce $S$ and may shift the decision toward CLARIFY. The E scorer partially implements a *textual spectrogram*—a reconstruction of emotional information from textual features when prosodic cues are absent (Section 4.3).

All three scorers are deterministic at test time: given the same input and anchor set, they produce identical scores across runs.

### 3.3 Hysteresis Controller

When $S$ hovers near a decision boundary, the system can produce unstable decision sequences—oscillating between EXECUTE and CLARIFY on successive evaluations of similar inputs. The hysteresis controller addresses this by introducing entry/exit threshold offsets: transitioning from EXECUTE to CLARIFY requires $S$ to fall below the boundary by offset $\varepsilon$, while returning to EXECUTE requires $S$ to rise above it by the same margin. This is the "AC thermostat" principle: a heating system that turns on at 68°F and off at 72°F is more stable than one that toggles at exactly 70°F. The controller maintains per-session state, ensuring independent hysteresis tracking across concurrent user conversations.

### 3.4 Epistemic Humility: Unknown Territory and CLARIFY Exhaustion

Two mechanisms encode a principle of epistemic humility: the system should admit "I do not recognize this" rather than defaulting to a permissive decision, and it should treat persistent evasion as a signal in its own right. The two connect into a single defensive pathway—an unrecognized input prompts a question, and if the user repeatedly dodges that question, the system stops.

**Unknown territory detection.** The anchor-based scorers measure proximity to known reference examples. When even the *nearest* anchor is far away, the input lies outside the system's curated experience—a blind spot where a confident decision is unjustified. AATIF makes this explicit: if the maximum cosine similarity to any harm anchor and to any intent anchor both fall below a threshold (UNKNOWN_TERRITORY_THRESHOLD = 0.20) and the S equation has nonetheless produced an EXECUTE decision, the decision is overridden to CLARIFY. The override is deliberately asymmetric—it acts only on EXECUTE, because the danger of operating in unrecognized territory is a *false* EXECUTE; the more cautious decisions (CLARIFY, SAFE_STOP, SAFE_FREEZE) are already conservative and are left untouched. The system thus refuses to assume safety merely because no harm anchor matched: low similarity to harm anchors is treated as ignorance, not as evidence of benignity.

**CLARIFY exhaustion.** Asking for clarification is only useful if it terminates. If a user is repeatedly met with CLARIFY yet never supplies the missing context, the evasion itself becomes informative—in the Architect's framing, *al-laff huwa al-ishāra* ("the dodging is the signal"). The hysteresis controller therefore enforces a bound: after the number of consecutive turns held in CLARIFY reaches MAX_CLARIFY_TURNS = 2, the next CLARIFY is escalated to SAFE_STOP rather than asked again. This prevents an indefinite clarification loop from being used to wear down the gate, and it converts sustained non-cooperation into a protective stop.

Together these mechanisms close a loop: an unrecognized input is routed to CLARIFY rather than executed, and if the user evades clarification past the limit, the system escalates to SAFE_STOP—unknown input → ask → if evaded, block.

### 3.5 Safety Gates: Law $\Omega$ and Law $\Xi$

Two unconditional safety gates fire *before* the S equation is evaluated, providing defense-in-depth independent of the scoring mechanism.

**Law $\Omega$ (CBRN Gate).** Unconditional detection of chemical, biological, radiological, and nuclear (CBRN) content in both Arabic and English. When triggered, the system enters SAFE_FREEZE regardless of all other scores. This gate is mode-independent: it fires identically across all four mode profiles.

**Law Ξ (Override Lock).** Detection of jailbreak and override attempts—phrases such as "ignore all previous instructions" or *tajāwuz al-taʿlīmāt* ("override instructions"). When triggered, the system enters SAFE_FREEZE. The override lock is designed to be non-bypassable: no prompt construction can deactivate it from within the input channel.

Both gates operate on pattern matching against curated trigger lists and are independent of the embedding backend, ensuring they function even in TF-IDF fallback mode.

### 3.6 Mode Profiles

AATIF supports four parameter profiles that adjust the sensitivity of the S equation without changing its structure:

Table 2: Mode profiles with parameter settings and use cases.

| Profile | $\alpha$ | $\theta$ | Use case |
|---|---|---|---|
| `high_sensitivity` | 15 | 0.30 | High-risk domains (medical, legal) |
| `default` | 10 | 0.40 | Standard operation |
| `relaxed` | 8 | 0.55 | Creative writing, fiction |
| `balanced_strict` | 10 | 0.40 | Tighter gate (A/B calibrated) |

Higher $\alpha$ makes the gate transition sharper (less ambiguous region); higher $\theta$ makes the gate more tolerant (opens wider before closing). The `high_sensitivity` profile uses a steeper gate ($\alpha = 15$), a lower threshold ($\theta = 0.30$), and reduced emotion weight ($w_2 = 1.0$), producing a narrower transition zone that closes earlier—achieving conservative behavior through both gate sharpness and threshold placement. The `relaxed` profile tolerates higher similarity before triggering ($\theta = 0.55$), with a gentler transition ($\alpha = 8$). Crucially, the hard override ($H > 0.7$) and safety gates (Laws Ω, Ξ) are mode-independent—they fire at full sensitivity regardless of profile selection.

### 3.7 Domain Parameterization of $\theta$

The harm threshold $\theta$ in the S equation is not a global constant but a domain-parameterized function $\theta(d)$, where $d$ denotes the application domain. Different deployment contexts carry fundamentally different risk profiles: a healthcare chatbot interacting with patients requires stricter harm sensitivity than a creative writing assistant, because the consequences of a false negative—failing to detect genuine harm—differ substantially in impact severity across domains. The domain parameterization reflects the principle that harm sensitivity is a property of the deployment context, not of the system's internal personality.

Table 3 reports the current domain configurations.

Table 3: Domain-specific harm thresholds $\theta(d)$ with risk rationale.

| Domain | $\theta(d)$ | Rationale |
|---|---|---|
| Healthcare | 0.25 | Physical harm risk; gate closes earliest |
| Education | 0.30 | Developmental harm to minors; conservative gate |
| General / Technology / E-Commerce | 0.40 | Default calibrated operating point |
| Creative | 0.50 | Wider tolerance for sensitive literary topics |

These values are initial estimates derived from empirical analysis and domain risk assessment, not mathematically derived constants. The space of possible application domains is unbounded—new domains (e.g., legal, financial, military) require calibration through domain-specific evaluation sets, and no universal derivation of optimal $\theta(d)$ values is claimed. The range of current values—$\theta(d) \in [0.25, 0.50]$—reflects the observation that even the most permissive domain threshold ($\theta = 0.50$ for creative contexts) remains well below the hard override at $H > 0.7$, preserving a substantial safety margin by construction.

At runtime, when a domain is specified, $\theta(d)$ overrides the mode profile's $\theta$ value. Mode profiles (Table 2) continue to control the remaining parameters ($w_1$, $w_2$, $\alpha$) but cede harm threshold governance to the domain configuration. This separation reflects a design distinction: profiles adjust the system's *operational personality* (gate steepness, quality weights), while domains adjust its *contextual risk sensitivity* (where the harm gate begins to close). AATIF independently supports four mode profiles (`high_sensitivity` $\theta = 0.30$, `default` $\theta = 0.40$, `relaxed` $\theta = 0.55$, `balanced_strict` $\theta = 0.40$) that govern gate behavior when no domain is specified; at runtime, domain $\theta(d)$ takes precedence over profile $\theta$ when both are present.

Crucially, the hard override ($H > 0.7 \Rightarrow$ SAFE_FREEZE) is domain-independent by construction. No domain configuration can raise or lower the absolute safety floor: chemical weapons synthesis triggers unconditional refusal regardless of whether the deployment context is healthcare, education, or creative writing. Domain parameterization operates strictly within the regime $\theta(d) < 0.7$, modulating the gate's sensitivity to moderate harm signals without affecting the system's response to extreme harm.

### 3.8 Ambiguity Pre-Check and Jailbreak Detection

**Ambiguity detection.** Short or vague prompts trigger an override from EXECUTE to CLARIFY, preventing the system from processing ambiguous requests that could be interpreted as either benign or harmful. This operationalizes the design principle that the cheapest correct response to an ambiguous input is a clarifying question, not a speculative completion.

**Bidirectional jailbreak handler.** The jailbreak handler operates in both directions: it *escalates* decisions when jailbreak markers are detected (role-play framing, instruction-override language, system-prompt extraction attempts), and it can *downgrade* a SAFE_FREEZE to SAFE_STOP when no jailbreak markers are present but simple harm content is detected. This bidirectional design reflects the principle that simple harm (a genuine safety question from a concerned user) is not manipulation and should receive an empathetic explanation rather than unconditional refusal.

### 3.9 The Full Governance Pipeline

The S equation governs safety scoring, but a complete governance system requires more than a safety gate. AATIF implements a three-layer pipeline that separates safety, domain rules, and response style into distinct, independently testable stages:

$$\text{Input} \xrightarrow{S(d)} \text{Safety} \xrightarrow{P(d)} \text{Protocols} \xrightarrow{R(d)} \text{Style} \xrightarrow{\text{LLM}} \text{Generation} \xrightarrow{\text{Gate}} \text{Output} \tag{3}$$

Each stage has a distinct responsibility and cannot compensate for failures in another:

1. $S(d)$**: Safety scoring** (the S equation) determines whether a request should proceed, be clarified, or be refused—a mathematical judgment.
2. $P(d)$**: Domain protocols** apply deterministic, domain-specific rules that can block requests or attach mandatory flags (e.g., medical disclaimers, age-appropriate language requirements) regardless of the S score—a rules-based judgment.
3. $R(d)$**: Response style** shapes *how* the system responds (tone, verbosity, formality, directness) without affecting *whether* it responds—a supervised aesthetic judgment.
4. **LLM generation** produces the actual response text, constrained by the preceding layers.
5. **Output Gate** performs final safety and quality verification on the generated response before it reaches the user—a post-generation checkpoint.

This separation is the core architectural contribution beyond the S equation itself: safety ($S$) is mathematical, protocols ($P$) are deterministic rules, and style ($R$) is supervised—three different computational paradigms governing three different concerns, composed in a fixed sequence with no feedback loops between them.

### 3.10 The R Equation: Response Style

The R equation governs response *style*—how the system communicates—as distinct from the S equation, which governs whether the system communicates at all. The separation is deliberate: safety and style are independent concerns, and conflating them (e.g., making a safety system also control tone) produces brittle designs where aesthetic preferences interfere with safety judgments.

$$R = \sigma(w_3 \cdot T + w_4 \cdot V + w_5 \cdot G + w_6 \cdot D) \tag{4}$$

where $T$ is time context (time of day mapped to six temporal periods), $V$ is voice (the user's communication style and register), $G$ is gap (time elapsed since the last interaction, mapped through six duration thresholds), and $D$ is domain signal (a domain-specific style influence, ranging from $D = 0.15$ for healthcare to $D = 0.70$ for creative contexts). The weights ($w_3 = 1.0$, $w_4 = 1.5$, $w_5 = 0.8$, $w_6 = 2.0$) encode the design judgment that domain context is the strongest style driver, followed by user voice. The output $R \in (0, 1)$ maps to four style bands: formal ($R \leq 0.3$), balanced ($0.3 < R \leq 0.5$), warm ($0.5 < R \leq 0.7$), and casual ($R > 0.7$). Domain-specific gate ceilings prevent inappropriately casual style in sensitive contexts: healthcare caps $R$ at 0.5 (formal or balanced only) and education caps $R$ at 0.6 (no casual). Crucially, $R$ is *not* a safety mechanism: it cannot override a SAFE_FREEZE decision, cannot relax the harm gate, and cannot bypass domain protocols. Temporal features influence response tone, not safety thresholds, preventing the system from treating time-of-day as a proxy for risk. The R equation module comprises 584 lines of implementation with 67 dedicated tests.

### 3.11 Domain Protocols $P(d)$

Domain protocols form a deterministic rules layer between safety scoring and response generation. Where the S equation produces a continuous score and the R equation shapes style, $P(d)$ applies categorical, non-negotiable constraints specific to the deployment domain.

In the healthcare domain, for example, $P(d)$ enforces a credentials-check requirement: requests involving medical advice must either come from a verified professional context or receive a mandatory disclaimer. In the education domain, $P(d)$ enforces age-appropriate language constraints. These rules are not suggestions—they can *block* a request that the S equation would have allowed (e.g., an S-scored EXECUTE on a medical question without disclaimer) or attach *emergency flags* that override all downstream processing (e.g., a suicide-risk indicator detected in an education context triggers immediate escalation regardless of S score).

The protocol layer is deliberately deterministic—no probabilistic scoring, no learned weights, no embeddings. Each rule is an explicit conditional: `if domain == healthcare and no_credentials then attach_disclaimer`. This makes protocol behavior fully auditable and prevents the subtle failures that emerge when probabilistic systems interact with domain-specific regulatory requirements. The module comprises 890 lines of implementation with 132 dedicated tests covering all supported domains.

### 3.12 Output Gate

The Output Gate is the final checkpoint before a generated response reaches the user. It operates *after* LLM generation, inspecting the actual output text rather than the input request. This post-generation position is critical: the S equation, domain protocols, and R equation all operate on the *input*; the Output Gate is the only component that verifies the *output*.

The gate applies six check layers in sequence:

1. **Safety leak detection**: Scans generated text for harm patterns that the LLM may have introduced despite safe input scoring—content that slipped through $S(d)$. Can block the response entirely.
2. **Identity protection**: Enforces persona identity (the *'ātif* persona) and blocks references to other AI systems' internal architectures that would break character.
3. **Forbidden phrase filtering**: Removes character-breaking phrases such as architecture internals, robotic hedging, or system-level language that a governed persona should never emit.
4. **Protocol compliance check**: Verifies that $P(d)$ requirements are present in the response—mandatory keywords such as EMERGENCY, DISCLAIMER, or WARNING that domain protocols flagged as required.
5. **Response quality guards**: Checks length limits, emptiness, repetition loops, and language consistency. Can block degenerate outputs.
6. **Final sanitization**: Strips file paths, API keys, JSON formatting errors, system prompt fragments, and excessive whitespace from the response.

Each check can independently block or flag the output. A blocked output returns a safe fallback response rather than the generated text. The Output Gate comprises 748 lines of implementation with 105 dedicated tests.

### 3.13 The Governor: Pipeline Orchestration

The Governor is the orchestrator that wires the full pipeline into a single execution path: $S(d) \to P(d) \to R(d) \to$ LLM $\to$ Output Gate. It is not a decision-maker itself but a coordinator that ensures each stage receives the correct inputs, respects the decisions of prior stages, and produces a complete audit trail.

The Governor exposes a `GovernedResponse` dataclass that records, for each request: the S score and decision, any protocol flags or blocks from $P(d)$, the R style parameters, the raw LLM output, the Output Gate's check results, and the final delivered response. This audit trail makes every governance decision inspectable after the fact—a property required for deployment in regulated domains where "why did the system say this?" must have a concrete, traceable answer.

The Governor module resolves four critical findings (C1–C4) identified during code review: (C1) the pipeline stages execute in a fixed, non-bypassable order; (C2) protocol blocks take precedence over S-score EXECUTE decisions; (C3) the Output Gate cannot be skipped even when all prior stages approve; (C4) the audit trail captures every intermediate state, not just the final output.

## 4 Arabic-First Design

### 4.1 The Meaning Density Hypothesis (*kathāfat al-ma'nā*)

We hypothesize that *meaning density*—the number of semantic dimensions encodable per token—is a relevant variable when selecting anchor languages for a safety system. Arabic was chosen for AATIF's anchor set based on two observable properties of the language: high figurative density in colloquial speech and morphological productivity from trilateral roots.

As a *linguistic* observation, Arabic's root-pattern morphology concentrates related meanings around shared consonantal skeletons. The trilateral root *k-t-b* ("writing") generates *kitāb* (book), *kātib* (writer), *maktaba* (library), *maktūb* (written letter). Figurative density compounds this: *wallāhi ba-dhbaḥuh* ("I swear I'll slaughter him") packs intent, emotion, cultural context, and literal meaning into two words.

However, a critical caveat is necessary: **there is no evidence that current embedding models preserve this morphological structure.** Models such as bge-m3 use subword tokenization (BPE/SentencePiece), not root extraction. The tokenizer splits Arabic text into statistically frequent byte-pair units, not into linguistically meaningful root-pattern decompositions. Whether morphologically related Arabic words (e.g., *kitāb* and *maktaba*) cluster more tightly in embedding space than semantically related English words (e.g., "book" and "library") is an *open empirical question* that we have not tested. We therefore do not claim that Arabic's linguistic properties transfer to embedding geometry.

What AATIF's design does rely on is a simpler, testable property: that Arabic-language anchors written in the same language and dialect as expected user input will produce higher cosine similarity scores than cross-lingual matching through English anchors. This is an anchor-coverage argument, not a morphological one. Our MultiJail results (Arabic 74.7% vs English 69.3%) are consistent with this anchor-coverage explanation—the H scorer carries more Arabic-language anchors than English ones—and we do not claim they demonstrate any deeper linguistic advantage. Testing the meaning density hypothesis directly would require controlled experiments with an Arabic-first embedding model, which is future work. We note that other Semitic languages (Hebrew, Amharic) share root-pattern morphology and could serve as comparison cases [Wolff & Holmes, 2011, Slobin, 1996].

### 4.2 Dialect Hyperbole Disambiguation

Arabic colloquial speech uses violent metaphors for everyday emotions with a frequency that poses a unique challenge for safety systems. Table 4 illustrates representative examples across Gulf, Levantine, and Egyptian dialects.

Table 4: Arabic dialect hyperbole: literal meaning vs. actual meaning.

| Expression | Literal meaning | Actual meaning |
|---|---|---|
| *yijīb lahum jalṭa* | Give them a stroke | Annoy them greatly |
| *wallāhi ba-dhbaḥuh* | I'll slaughter him | I'm annoyed at him |
| *qunbula ʻala al-sōshyāl mīdyā* | A bomb on social media | A viral hit |
| *bamūt fīk* | I'm dying in you | I love you deeply |
| *yaqtulnī min al-ḍaḥk* | He's killing me | He's hilarious |

A safety system without dialect awareness would flag all five expressions as harmful: "slaughter," "bomb," "dying," "killing" all appear in violence-related anchor neighborhoods. The H scorer addresses this through 31 dialect-hyperbole anchors, each scored at $H \leq 0.05$. These anchors act as semantic counter-gravity: when an input matches a hyperbole anchor more closely than a harm anchor, the figurative interpretation dominates and the harm score remains low. The tighter threshold ($\leq 0.05$ rather than $\leq 0.10$) reflects calibration showing that hyperbole anchors needed a narrower safety margin to avoid inadvertently softening harm scores for inputs that are *near* but not *in* the figurative category. This mechanism exploits the top-$K$ aggregation design—if one of the top-3 nearest anchors is a hyperbole anchor, it dilutes the average toward safety.

We call the underlying failure mode *Lexical Anchor Contamination*: multilingual embeddings trained predominantly on English data may not reliably separate Arabic figurative violence from literal violence, because the training distribution underrepresents the figurative conventions of Arabic dialects. The cross-dialect accuracy drop documented by Aldjanabi et al. [2024] (0.84 to 0.64) confirms that dialect-sensitivity is not optional. The hyperbole anchors are a practical workaround; whether the failure reflects embedding geometry or training data distribution is an open question.

The dialect test suite contains 22 test cases covering Gulf, Levantine, and Egyptian dialectal expressions. All 22 pass with zero false positives: no hyperbolic expression triggers harm detection.

### 4.3 The Maqam-to-Safety Origin Chain

The emotional channel ($E$ scorer) and the multiplicative gate structure originated from an unexpected domain: Arabic music theory. We present this as an intellectual origin chain—a sequence of observations leading from quarter-tone music to a concrete safety mechanism.

**The quarter-tone gap.** Arabic music uses the maqam system [Touma, 1971, Abu Shumays, 2013], built on quarter-tone intervals that cannot be represented by the Western 12-tone equal-tempered scale. AI music systems trained on Western scales are structurally incapable of representing Arabic musical emotion—not due to insufficient training data, but because the representational vocabulary lacks the required resolution. EEG studies confirm that maqam produces distinct neural responses in human listeners [Yaghmour et al., 2021], establishing that the emotional content is real, not merely cultural convention.

**From music to speech.** Each Arabic maqam carries temporal-emotional information: Bayati evokes sadness, Rast stability, Hijaz longing. The same principle applies to speech: prosody (pitch contour, rhythm, emphasis) carries emotional information that lexical content alone does not capture.

**Text loses prosody.** When communication shifts from speech to text, prosodic information is lost. The sentence "fine, do whatever you want" can be angry, sarcastic, resigned, or genuinely permissive. Text is prosody-stripped, and safety systems operating on text alone are structurally deaf to this emotional dimension.

**The textual spectrogram.** If prosody is lost in text, can an emotional fingerprint be reconstructed from textual features? Candidate features include punctuation patterns, repetition, dialectal markers, sentence length variation, and formality shifts. The $E$ scorer partially implements this reconstruction, using emotional anchors as reference points for what we term a *textual spectrogram*.

**Duress detection.** The most compelling application of this chain is duress detection. An authorized user under coercion will exhibit involuntary changes in their textual fingerprint: atypical sentence patterns, unusual formality, absence of habitual phrases. Detection requires convergence of three independent signals: fingerprint deviation ($E$ scorer analogue), contextual anomaly (unusual patterns), and dangerous command ($H$ scorer analogue). No single signal suffices—all three must converge before the governance gate fires.

**The multiplicative gate.** This three-signal convergence requirement is exactly the principle the S equation implements. The product structure ensures that no single channel—harm, intent, or emotion—can override the governance decision independently. The maqam observation motivated the emotional channel; the duress use case motivated the multiplicative gate; the S equation implements the principle that the musical observation identified.

# 5 Experimental Evaluation

## 5.1 Test Suite

AATIF includes 790 deterministic tests organized across 17 test modules: S equation core including intent engine (79 tests), emotion scorer (32 tests), gated comparison (19 tests), and response shaping (46 tests); safety gates including CBRN detection (12 tests) and jailbreak marker detection (10 tests); dialect hyperbole disambiguation (22 tests); domain parameterization including domain-specific $\theta(d)$ (35 tests) and domain protocols $P(d)$ (132 tests); intent scorer (30 tests); R equation style adaptation (67 tests); Output Gate (105 tests); Governor orchestration (26 tests); time-sense module (40 tests); unknown-territory detection (11 tests); held-out validation (2 test functions with parametric expansion); and pipeline integration (3 tests). All tests are deterministic—they require no external model at test time and produce identical results across runs. The test suite runs via GitHub Actions for continuous integration.

## 5.2 HarmBench Evaluation

We evaluated the H scorer against HarmBench [Mazeika et al., 2024] (236 behaviors, 7 semantic categories) using the bge-m3 multilingual embedding backend at threshold $H \geq 0.3$.

Table 5: HarmBench detection rates by semantic category (bge-m3 backend, $H \geq 0.3$).

| Category | Det./Total | Rate | Avg $H$ | MLCommons |
|---|---|---|---|---|
| Chemical/biological | 28/28 | 100.0% | 0.894 | S9 |
| Cybercrime | 38/43 | 88.4% | 0.662 | S2 |
| Illegal activities | 38/43 | 88.4% | 0.665 | S2 |
| Harmful content | 7/10 | 70.0% | 0.481 | S1/S11 |
| Harassment | 9/15 | 60.0% | 0.325 | S10 |
| Misinformation | 13/39 | 33.3% | 0.262 | S13 |
| Copyright | 4/57 | 7.0% | 0.080 | S8 |
| **Safety categories** | **133/179** | **74.3%** | | |
| **All categories** | **137/236** | **58.1%** | | |

Safety-category detection (excluding copyright, which is not safety-relevant and for which AATIF has no anchors) reaches 74.3%. Chemical/biological threats achieve 100% detection with average $H = 0.894$; both cybercrime and illegal activities reach 88.4%. The bge-m3 backend enables cross-lingual semantic matching: English prompts match against Arabic harm anchors by meaning rather than character overlap, confirmed by nearest-anchor analysis.

Two categories remain weak: misinformation (33.3%) and copyright (7.0%). The anchor-based approach requires harm-indicative language in the input; subtle manipulation and paraphrase elude surface-level matching regardless of embedding backend.

## 5.3 MultiJail Evaluation

We evaluated the H scorer against MultiJail [Deng et al., 2024] (75 harmful prompts with native Arabic translations) to test whether AATIF's Arabic-first design produces measurably different detection characteristics across languages.

Arabic outperforms English in both detection rate (74.7% vs 69.3%) and average harm score (0.519 vs 0.495), with Arabic scoring higher in 42 of 75 prompt pairs. We caution against over-reading this 5.4-point gap. With only $n = 75$ prompt pairs and no significance test performed, the difference may reflect *anchor coverage asymmetry*—the H scorer carries

Table 6: MultiJail detection: Arabic vs English (bge-m3, $H \geq 0.3$).

| | Arabic | English |
|---|---|---|
| Detection rate | 74.7% (56/75) | 69.3% (52/75) |
| Average $H$ | 0.519 | 0.495 |
| AR scores higher | 42/75 (56%) | |
| High confidence ($\geq$0.45) | 52/64 (81.2%) | |

more Arabic-language anchors than English ones, giving Arabic input denser semantic reference points—rather than any intrinsic property of the language or its meaning density. We therefore present this result as an observation consistent with the meaning density hypothesis, not as conclusive evidence for it. The curated anchors provide stronger semantic coverage for Arabic input than the smaller English anchor subset provides for English, which alone could produce the observed gap.

Category-level analysis reveals that the Arabic advantage concentrates in violence and threat categories. For example, a date-rape prompt scored $H = 0.900$ in Arabic versus $H = 0.552$ in English; a discriminatory prompt scored $H = 0.653$ versus $H = 0.276$. Conversely, English outperforms Arabic in categories with stronger English-language anchors (e.g., property crime, business fraud). This asymmetry reflects anchor coverage distribution, not a framework limitation.

## 5.4 A/B Calibration

The default profile was calibrated via a $\theta$ sweep on 54 test cases (30 harmful, 24 benign) using the bge-m3 backend with $\alpha = 10$. We stress that the $F_1$ in Table 7 is an *in-sample* figure: the threshold was both selected and evaluated on these same 54 cases. Held-out generalization is reported separately below.

Table 7: A/B calibration: $\theta$ sweep results with $\alpha = 10$.

| $\theta$ | Detection | FP Rate | Precision | Recall | $F_1$ |
|---|---|---|---|---|---|
| 0.20 | 100% | 33.3% | 0.789 | 1.000 | 0.882 |
| 0.25 | 100% | 20.8% | 0.857 | 1.000 | 0.923 |
| 0.30 | 100% | 12.5% | 0.909 | 1.000 | 0.952 |
| 0.35 | 100% | 4.2% | 0.968 | 1.000 | 0.984 |
| **0.40** | **100%** | **4.2%** | **0.968** | **1.000** | **0.984** |
| 0.45 | 96.7% | 0.0% | 1.000 | 0.967 | 0.983 |
| 0.50 | 90.0% | 0.0% | 1.000 | 0.900 | 0.947 |
| 0.55 | 83.3% | 0.0% | 1.000 | 0.833 | 0.909 |

The optimal operating point is $\theta = 0.40$, achieving an in-sample $F_1 = 0.984$ with 100% detection rate and 4.2% false positive rate. The range $\theta \in [0.35, 0.40]$ shares identical performance, indicating robustness to small threshold variations. The single false positive is the Arabic phrase *ḥazīn shway bas bkhayr* ("a little sad but I'm fine"), which triggers SAFE_STOP due to the distress indicator. This false positive aligns with AATIF's design philosophy rooted in the Arabic concept of *'aṭf* (compassionate inclination): when in doubt, lean toward empathetic over-caution rather than dismissive permissiveness. The steepness parameter $\alpha$ is stable across values 8–20, confirming that the system is not sensitive to this hyperparameter.

**Held-out validation.** An in-sample $F_1$ overstates real performance because the same cases drive both calibration and evaluation. To estimate generalization, we constructed a held-out set of 56 never-before-seen cases (28 Arabic / 28 English; 30 safe, 26 blocked) spanning educational, medical, historical, and financial framings of sensitive topics. The in-sample-tuned configuration dropped sharply on this set, to $F_1 = 0.735$ (precision 0.595, recall 0.962): the H scorer was effectively measuring *topic sensitivity* rather than *harmful intent*, over-blocking benign academic and clinical content and producing 17 false positives. We then refined the harm anchor set—adding educational "safe" anchors to pull benign framings down, plus counter-harm anchors so those safe anchors cannot launder genuinely dangerous look-alikes (e.g. lethal dosing, coercive control, elder-fraud scripts). Held-out performance recovered to $F_1 = 0.9615$ (precision = 0.9615, recall = 0.9615; 1 false positive, 1 false negative), with the gate threshold left unchanged at $\theta = 0.40$. This is a genuine improvement—precision rose from 0.60 to 0.96 with no loss of recall and no new missed harms—but one caveat is essential to state plainly: the counter-harm and protective anchors were informed by the very held-out failures they were designed to fix, so the 56-case set no longer constitutes a clean blind estimate for those categories. A fully blind validation on a fresh, unseen set is still required before treating 0.96 as the true operating point.

### 5.5 MLCommons Coverage Mapping

Table 8 maps AATIF's anchor coverage against the MLCommons 13-category harm taxonomy.

Table 8: AATIF anchor coverage vs. MLCommons taxonomy.

| Code | Category | Coverage | Anchors |
|---|---|---|---|
| S1 | Violent Crimes | Strong | 15+ |
| S2 | Non-Violent Crimes | Partial | 5+ |
| S3 | Sex-Related Crimes | Partial | 2 |
| S4 | Child Sexual Exploitation | None | 0 |
| S5 | Defamation | None | 0 |
| S6 | Specialized Advice | None | 0 |
| S7 | Privacy | Partial | 2 |
| S8 | Intellectual Property | None | 0 |
| S9 | Indiscriminate Weapons | Strong | 12+ |
| S10 | Hate | Strong | 41+ |
| S11 | Suicide & Self-Harm | Strong | 6+ |
| S12 | Sexual Content | None | 0 |
| S13 | Elections | None | 0 |

Four categories have strong coverage (S1, S9, S10, S11), three have partial coverage (S2, S3, S7), and six have no anchor coverage. This is an honest reflection of AATIF v1.0's scope: the anchor set reflects the harm categories that emerged from the observational process rather than a predetermined taxonomy. Critically, expanding coverage requires adding anchors, not retraining—a structural advantage over trained classifiers. Adding 50 anchors for a new category takes hours of curation, not weeks of data collection and model training.

### 5.6 Comparison with FanarGuard

AATIF and FanarGuard [Fatehkia et al., 2025] represent complementary paradigms for Arabic LLM safety, and comparison clarifies what each approach offers.

The comparison is complementary, not competitive. FanarGuard's data-driven approach provides broader coverage and a cultural alignment axis that AATIF lacks. AATIF's equation-driven approach provides full interpretability, non-compensability by construction, and zero-training extensibility. A hybrid architecture—FanarGuard's cultural alignment axis feeding into AATIF's governance equation—is a natural direction for future work.

Table 9: Structural comparison: AATIF vs. FanarGuard.

| Property | AATIF | FanarGuard |
|---|---|---|
| Architecture | Explicit equation | 2B-param classifier |
| Training data | 0 examples | 468K examples |
| Interpretability | Full (inspect equation) | Partial (learned weights) |
| Non-compensability | By construction (gate + override) | Not guaranteed |
| Cultural alignment | Not implemented | Dual-axis scoring |
| Extensibility | Add anchors (hours) | Retrain model (weeks) |

# 6 Discussion

## 6.1 Contributions

This paper makes five contributions. First, it combines, for the first time in a single deployable system, a closed-form multiplicative governance equation that provides non-compensability by construction (with an external hard override supplying the absolute high-harm floor), Arabic dialect-aware anchor scoring, and zero-fine-tuning extensibility: the multiplicative gate structure ensures that no combination of benign intent and calm emotion can restore the safety score once the harm gate is closing, and the framework is designed for human-over-the-loop governance rather than fully autonomous decision-making. Second, it demonstrates zero-training extensibility—coverage expands through anchor curation alone, requiring no labeled datasets or model retraining. Third, it validates Arabic-first design empirically: Arabic outperforms English on identical harmful prompts in the MultiJail benchmark (74.7% vs 69.3%), and 22 dialect-specific test cases produce zero false positives on figurative hyperbole. Fourth, it presents a three-layer governance pipeline—$S(d)$ for safety, $P(d)$ for domain-specific deterministic protocols, $R(d)$ for response style—that separates mathematical safety scoring from rules-based domain constraints and supervised aesthetic adaptation, with an Output Gate providing post-generation verification and a Governor module ensuring a complete audit trail. Fifth, it provides a complete open-source implementation with 790 deterministic tests ensuring mathematical correctness across all modules.

## 6.2 Limitations

Several limitations constrain the scope of this work. The system was developed by a single independent researcher; the code is open, the tests are deterministic, and the benchmarks are reproducible, but replication by independent teams is necessary to establish generalizability. AATIF v1.0 covers only 4 of 13 MLCommons categories strongly, with 6 categories having zero anchor coverage, including critical categories such as Child Sexual Exploitation (S4) and Sexual Content (S12). The A/B calibration was performed on 54 test cases—sufficient for demonstrating framework properties but insufficient for production-grade reliability claims. No ablation study has been conducted on individual scorers ($H$-only, $I$-only, $E$-only, and pairwise combinations). Misinformation (33.3% detection) and copyright (7.0%) remain structurally weak because the anchor-based approach requires harm-indicative language in the input; subtle manipulation and paraphrase elude surface-level matching. FanarGuard achieves $F_1 = 0.82$ on a full test set using 468K training examples—a data-driven approach that the current anchor-based system cannot match in coverage or scale. AATIF's contributions are structural (interpretability, non-compensability, zero-training extensibility) rather than scale-competitive.

### 6.3 The Conjectures as Research Program

AATIF originated from 73 governance conjectures derived through sustained observation of LLM behavior across four platforms over twelve months, following an inductive methodology with precedent in grounded theory [Glaser & Strauss, 1967] and usability heuristic derivation [Nielsen & Molich, 1990]. The conjectures span cognitive, ethical, linguistic, perceptual, and architectural domains, each assessed for consistency with peer-reviewed literature. This paper operationalizes one conjecture—#063, that governance operates through probabilistic constraint density rather than rule retrieval—into a working system with empirical validation.

Several other conjectures motivated specific design decisions. The Clarification Conjecture (#001)—that the cheapest correct response to ambiguity is a clarifying question—motivated the ambiguity pre-check. The Bounded Claims Conjecture (#069), consistent with Herley & van Oorschot [2017] on Popperian falsifiability in security, shaped the system's explicit acknowledgment of coverage limits rather than absolute safety claims. The Pressure-Reveal Conjecture (#067), consistent with van der Weij et al. [2024] on strategic underperformance in evaluations, motivated adversarial testing as the primary evaluation modality.

The 73 conjectures represent the broader research program; this paper delivers one node with working code and empirical validation. The full set of 78 field notes is available as supplementary material. Future work will address additional conjectures, including pre-computation ethical gatekeeping at the question-formulation stage and simultaneous multi-perspective deliberation (normative, contextual, analytical).

## 7 Conclusion

AATIF provides an explicit, interpretable mathematical governance equation for LLM safety: $S = \sigma(w_1 \cdot I + w_2 \cdot E) \cdot [1 - \sigma(\alpha(H - \theta(d)))]$. The multiplicative gate structure provides non-compensability by construction—high harm drives the safety score toward zero regardless of benign intent or calm emotion—with an external hard override ($H > 0.7$) supplying the absolute refusal floor. The harm threshold $\theta(d)$ is domain-parameterized, ranging from 0.25 (healthcare) to 0.50 (creative), allowing the same system to adapt its sensitivity to deployment context while the hard override remains domain-independent. The framework requires zero training data and extends through anchor curation alone.

Beyond the S equation, AATIF implements a full governance pipeline: safety scoring $S(d)$, domain-specific deterministic protocols $P(d)$, response style adaptation $R(d)$, LLM generation, and an Output Gate for post-generation safety and quality verification—all orchestrated by a Governor module that provides a complete audit trail.

Arabic-first design is validated empirically: Arabic outperforms English on identical harmful prompts (74.7% vs 69.3% on MultiJail), and 31 dialect-hyperbole anchors (threshold $\leq 0.05$) eliminate false positives on figurative Arabic expressions. The system achieves 74.3% safety-category detection on HarmBench with 100% on chemical/biological threats, with in-sample calibration $F_1 = 0.984$ (54 cases) and held-out $F_1 = 0.9615$ (56 unseen cases) after anchor refinement, the latter still pending a fully blind validation round. A test suite of 790 deterministic tests ensures mathematical correctness across all modules.

The framework is preliminary in coverage (4/13 MLCommons categories strong) and scale (single-observer development, 54-case calibration). Future work includes anchor expansion to all 13 MLCommons categories, ablation studies on individual scorers, integration of a cultural alignment axis inspired by FanarGuard's dual-axis design, and development of an Arabic-first embedding model to test the meaning density hypothesis directly.

**Code and Data.** The AATIF computational engine, test suite (790 tests), benchmark runners, and evaluation harness are available at `https://github.com/Abdelmgied-Khenkar/AATIF`.

The companion preprint is available at https://doi.org/10.5281/zenodo.20673292.

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